Unit: Unit 3: Functions
Topic: Inverse Relations and Functions

Name: ____________
Date: ________

Challenge (Habanero)

Q1. A recipe for cake requires $2x$ cups of flour and $3x$ cups of sugar. If $f(x)$ represents the amount of flour and $g(x)$ represents the amount of sugar, what is the inverse of the relation $f(g(x))$?

(A) $\frac{x}{6}$
(B) $\frac{x}{3}$
(C) $\frac{x}{2}$
(D) x
(E) $2x$

Q2. A building's height is $5x$ meters, and its shadow is $2x$ meters. If the function $h(x)$ represents the height and $s(x)$ represents the shadow length, what is $h^{-1}(s(x))$?

(A) $\frac{x}{10}$
(B) $\frac{x}{5}$
(C) $\frac{x}{2}$
(D) x
(E) $2x$

Q3. If $f(x) = 2x+1$, what is the equation of its inverse function $f^{-1}(x)$?

(A) $\frac{x-1}{2}$
(B) $\frac{x+1}{2}$
(C) $\frac{x-2}{2}$
(D) $\frac{x+2}{2}$
(E) $x-2$

Q4. A construction company uses the function $c(x) = 3x-2$ to calculate the cost of building a house, where x is the number of rooms. What is $c^{-1}(x)$?

(A) $\frac{x+2}{3}$
(B) $\frac{x-2}{3}$
(C) $\frac{x+1}{2}$
(D) $\frac{x-1}{2}$
(E) $x+2$

Q5. If $f(x) = x^2+2$, does $f(x)$ have an inverse function?

(A) Yes
(B) No
(C) Maybe
(D) Only for $x > 0$
(E) Only for $x < 0$

Q6. The volume V of a rectangular prism is given by $V = lwh$, where l is the length, w is the width, and h is the height. If $l = 2w$ and $h = 3w$, what is the inverse of the relation $V(w)$?

(A) $\frac{x}{12}$
(B) $\frac{x}{6}$
(C) $\sqrt[3]{\frac{x}{6}}$
(D) $\sqrt{\frac{x}{6}}$
(E) $\frac{x}{2}$

Q7. A chef uses the function $t(x) = 2x-5$ to determine the cooking time for a dish, where x is the number of servings. What is $t^{-1}(x)$?

(A) $\frac{x+5}{2}$
(B) $\frac{x-5}{2}$
(C) $\frac{x+2}{2}$
(D) $\frac{x-2}{2}$
(E) $x+5$

Q8. If $f(x) = x^2-4$, does $f(x)$ have an inverse function?

(A) Yes
(B) No
(C) Maybe
(D) Only for $x > 0$
(E) Only for $x < 0$

Open Ended Questions (FRQ)

Q9. A contractor uses the function $c(x) = 2x+5$ to calculate the cost of building a house, where x is the number of rooms. Find the inverse of $c(x)$ and interpret its meaning in the context of the problem. Show all work and explain your answer.

Q10. A baker uses the function $s(x) = 3x-2$ to determine the amount of sugar needed for a recipe, where x is the number of servings. Find the inverse of $s(x)$ and use it to find the number of servings that require 5 cups of sugar. Show all work and explain your answer.

Chef's Tips

- Read each question carefully
- Use the given equations to find the inverse functions
- Show all work and explain your answers